

Chainstack: Making blockchain work for enterprises and developers with Google Cloud

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ABOUT CHAINSTACK



With Google Cloud, Chainstack is positioned to become a de facto platform for enterprises to build decentralized applications and smart contracts. Its partners, Chainstack, help enterprises (

About Chainstack

Founded in April 2018, Chainstack delivers what it describes as a “control panel for blockchains.” The Chainstack platform—that currently operates across multiple clouds, supports a range of protocols, and is delivered as a service—enables enterprises and developers to rapidly build, deploy, and manage decentralized networks and services.

Industries: Technology

potential of the blockchain to secure and simplify transactions, build collaborative relationships and enhance applications

Location: Singapore

(<https://cloud.google.com/bigquery/>), Looker

Google Cloud (<https://cloud.google.com/>)
<https://cloud.google.com/about/data-engineering/>

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

<https://cloud.google.com/appengine/>

App Engine (<https://cloud.google.com/appengine/>)

Cloud SQL (<https://cloud.google.com/sql/>)

BigQuery (<https://cloud.google.com/bigquery/>)

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Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Anthos (<https://cloud.google.com/anthos/>)

Cloud SQL for PostgreSQL

(<https://cloud.google.com/sql/docs/postgres/>)

Google Cloud results

- Minimizes management load involved in running 500+ nodes
- Gains opportunity to run seamlessly across on-premises and cloud environments
- Deploys new nodes in less than three minutes

Deliver “blockchain” control for enterprises and deploy with Google Cloud

After emerging as the enabler of cryptocurrencies such as bitcoin, blockchain is establishing itself as

among the most important recent developments in technology. Blockchain networks allow transactions and information to be recorded and distributed in a peer-to-peer—rather than centralized—model. This minimizes the risk of third parties tampering with the data. Financial institutions and other businesses worldwide are operating blockchain networks, with the Financial Times reporting

(<https://www.ft.com/content/6363036a-d597-11e9-a0bd-ab8ec6435630>)

that 320 banks have signed up to JP Morgan's blockchain-based Interbank Information Network.

Blockchain has opened up a range of opportunities for entrepreneurs and startups—including Singapore—headquartered Chainstack (<https://chainstack.com/>). Founded in April 2018, Chainstack employs about 20 people to deliver a platform it describes as a “control panel for blockchains.” The Chainstack platform—that currently operates across multiple clouds, supports a range of protocols, and is delivered as a service—enables enterprises and developers to rapidly build, deploy, and manage decentralized networks and services.

Addressing the gap

Chief Executive Officer Laurent Dedenis believes Chainstack addresses the current gap between the relative immaturity of blockchain—manifested in lack of documentation and expertise and near-constant change—and its use within the enterprise. As the blockchain matures, Chainstack and its partners will help enterprises exploit its potential to transform transactions, enhance applications, and

build collaborative relationships between customers, partners, vendors, and other parties.

“Some applications will be better, faster, cheaper and provide more features in a decentralized environment,” says Dedenis. “Further, blockchain reduces costs and improves visibility and traceability of transactions. It does this by passing a single transaction over a network to multiple participants.”

Chainstack also provides consistency and ease of use at a time when blockchain protocols—the systems and methods used to validate transactions within the blockchain—are relatively immature. “We make it easy for partners like Deloitte and Fujitsu to identify opportunities to run processes over a decentralized network,” says Dedenis.

After launching its platform in the first half of 2019, Chainstack has now signed up partners in 15 countries including the United States.

“Ultimately, we are an infrastructure provider that enables enterprises to run blockchain in an enterprise way—with the right security, reliability, and monitoring,” says Dedenis.

Chainstack knew from inception that cloud would be the cornerstone of its platform. “While a lot of enterprises still operate their own data centers and on-premises equipment, most new projects and new companies start in the cloud,” says Eugene Aseev, Chainstack Chief Technology Officer. “We wanted to merge the concept of blockchain—a new way of

building networks and applications—with the cloud, which is the standard for building infrastructure.”

The decentralized nature of the blockchain— involving different locations and service providers— meant Chainstack needed to operate across multiple clouds. However, the business needed one cloud to act as the cornerstone of its architecture. With team members already having established relationships with Google Cloud account representatives and technical experts, Chainstack was well aware of the reliability, performance, and innovation of Google Cloud services.

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Executive Officer,

Google Cloud an easy choice

“Google Cloud was an easy choice for us because we were fascinated by the opportunities the platform presented to us as an engineering team,” says Dedenis. “In particular, we needed to use container orchestration—we were on the way to the 500+ nodes we run now, and it would have been difficult to manage them with traditional virtual machines or servers.” Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)—a managed, production-ready environment for deploying containerized applications—met the business’s needs and ensured it did not have to dedicate valuable technical resources to administration tasks.

“This complemented our engineering infrastructure strategy of using only managed services,” says Dedenis. “Our team is very flexible and agile, and we don’t want to have to commit them to managing servers.

“In addition, Google’s granular pay-per-use model aligns closely with our charging model.”

Chainstack developed and ran its platform in App Engine (<https://cloud.google.com/appengine/>) before moving to Google Kubernetes Engine to run customer deployments and blockchain protocols.

The business operates Stackdriver

(<https://cloud.google.com/stackdriver/>) to monitor

performance and obtain diagnostic data; BigQuery

(<https://cloud.google.com/bigquery/>) to run an analytics

data warehouse; Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

to provide analytics visualization; Cloud SQL

(<https://cloud.google.com/sql/>) to manage and

maintain SQL databases; and Cloud SQL for

PostgreSQL

(<https://cloud.google.com/sql/docs/postgres/>) to run its

PostgreSQL database.

“Anthos opens
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businesses in
healthcare,
banking, and
finance and
similarly heavily
regulated
industries to use
the blockchain.”

—Eugene Aseev, Chief
Technology Officer.
Chainstack

Evaluating Anthos

Chainstack is now looking into the potential of Anthos (<https://cloud.google.com/anthos/>)—which allows businesses to build and manage hybrid applications on-premises or in the cloud—to extend its capabilities. “This would be a game-changer for us—not only could we support cloud, but on-premises environments for businesses,” says Dedenis. “Anthos effectively provides a seamless transition from cloud to on-premises to deliver the same experience through the same interface.”

According to Aseev, this would enable Chainstack to offer its platform to businesses that operate in regulated industries and effectively have to use at least some on-premises storage and servers. “Anthos opens up new ways for businesses in healthcare, banking, and finance and similarly heavily regulated industries to use the blockchain,” he says.

Chainstack is now looking for a partner to develop a solution comprising an infrastructure layer—an on-premises network or nodes—based on Anthos, the Chainstack platform to orchestrate the blockchain network, and an application layer built on top of the blockchain protocol.

With Google Cloud, Chainstack can deploy nodes in less than three minutes and provide scale and

availability for blockchain networks—including the programmable Ethereum blockchain, one of the largest and most active blockchain communities in the world.

“We look forward to working with Google Cloud and partners to help enterprises make the blockchain frictionless and unleash the technology’s full potential.”

—Laurent Dedenis, Chief
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Chainstack

Eyeing expansion

Now well established, the business is eyeing expansion within the enterprise market. “We want to

become the de facto platform for people to come and build decentralized applications and smart contracts that execute code to fulfill a set purpose such as transferring value from one person to another,” says Dedenis. “As the market matures, consortia will form and blockchain networks will emerge and grow—our job will be to enable the deployment and management of these networks.

“We look forward to working with Google Cloud and partners to help enterprises make the blockchain frictionless and unleash the technology’s full potential.”

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business.